

statement can be made more precise by bringing in the machinery of obstruction theory, as explained in [Milnor & Stasheff 1974] or [VBKT].

Before deriving the Gysin sequence let us look at some examples of how it can be used to compute cup products.

Example 4D.5. Consider a bundle $S^{n-1} \rightarrow E \xrightarrow{p} B$ with E contractible, for example the bundle $S^1 \rightarrow S^\infty \rightarrow \mathbb{CP}^\infty$ or its real or quaternionic analogs. The long exact sequence of homotopy groups for the bundle shows that B is $(n-1)$ -connected. Thus if $n > 1$, B is simply-connected and we have a Gysin sequence for cohomology with $\mathbb{Z}$ coefficients. For $n = 1$ we take $\mathbb{Z}_2$ coefficients. If $n > 1$ then since E is contractible, the Gysin sequence implies that $H^i(B; \mathbb{Z}) = 0$ for $0 < i < n$ and that $\smile e: H^i(B; \mathbb{Z}) \rightarrow H^{i+n}(B; \mathbb{Z})$ is an isomorphism for $i \geq 0$. It follows that $H^*(B; \mathbb{Z})$ is the polynomial ring $\mathbb{Z}[e]$. When $n = 1$ the map $p^*: H^{n-1}(B; \mathbb{Z}_2) \rightarrow H^{n-1}(E; \mathbb{Z}_2)$ in the Gysin sequence is surjective, so we see that $\smile e: H^i(B; \mathbb{Z}_2) \rightarrow H^{i+n}(B; \mathbb{Z}_2)$ is again an isomorphism for all $i \geq 0$, and hence $H^*(B; \mathbb{Z}_2) \approx \mathbb{Z}_2[e]$. Thus the Gysin sequence gives a new derivation of the cup product structure in projective spaces. Also, since polynomial rings $\mathbb{Z}[e]$ are realizable as $H^*(X; \mathbb{Z})$ only when e has dimension 2 or 4, as we show in Corollary 4L.10, we can conclude that there exist bundles $S^{n-1} \rightarrow E \rightarrow B$ with E contractible only when n is 1, 2, or 4.

Example 4D.6. For the Grassmann manifold $G_n = G_n(\mathbb{R}^\infty)$ we have $\pi_1(G_n) \approx \pi_0 O(n) \approx \mathbb{Z}_2$, so the universal cover of G_n gives a bundle $S^0 \rightarrow \tilde{G}_n \rightarrow G_n$. One can view $\tilde{G}_n$ as the space of oriented n -planes in $\mathbb{R}^\infty$, which is obviously a 2-sheeted covering space of G_n , hence the universal cover since it is path-connected, being the quotient $V_n(\mathbb{R}^\infty)/SO(n)$ of the contractible space $V_n(\mathbb{R}^\infty)$. A portion of the Gysin sequence for the bundle $S^0 \rightarrow \tilde{G}_n \rightarrow G_n$ is $H^0(G_n; \mathbb{Z}_2) \xrightarrow{\smile e} H^1(G_n; \mathbb{Z}_2) \rightarrow H^1(\tilde{G}_n; \mathbb{Z}_2)$. This last group is zero since $\tilde{G}_n$ is simply-connected, and $H^1(G_n; \mathbb{Z}_2) \approx \mathbb{Z}_2$ since $H^*(G_n; \mathbb{Z}_2) \approx \mathbb{Z}_2[w_1, \dots, w_n]$ as we showed earlier in this section, so $e = w_1$ and the map $\smile e: H^*(G_n; \mathbb{Z}_2) \rightarrow H^*(G_n; \mathbb{Z}_2)$ is injective. The Gysin sequence then breaks up into short exact sequences $0 \rightarrow H^i(G_n; \mathbb{Z}_2) \xrightarrow{\smile e} H^{i+1}(G_n; \mathbb{Z}_2) \rightarrow H^{i+1}(\tilde{G}_n; \mathbb{Z}_2) \rightarrow 0$, from which it follows that $H^*(\tilde{G}_n; \mathbb{Z}_2)$ is the quotient ring $\mathbb{Z}_2[w_1, \dots, w_n]/(w_1) \approx \mathbb{Z}_2[w_2, \dots, w_n]$.

Example 4D.7. The complex analog of the bundle in the preceding example is a bundle $S^1 \rightarrow \tilde{G}_n(\mathbb{C}^\infty) \rightarrow G_n(\mathbb{C}^\infty)$ with $\tilde{G}_n(\mathbb{C}^\infty)$ 2-connected. This can be constructed in the following way. There is a determinant homomorphism $U(n) \rightarrow S^1$ with kernel $SU(n)$, the unitary matrices of determinant 1, so S^1 is the coset space $U(n)/SU(n)$, and by restricting the action of $U(n)$ on $V_n(\mathbb{C}^\infty)$ to $SU(n)$ we obtain the second row of the commutative diagram at the right. The second row is a fiber bundle by the usual argument of choosing continuously varying orthonormal bases in n -planes near a

$$\begin{array}{ccccc} U(n) & \longrightarrow & V_n(\mathbb{C}^\infty) & \longrightarrow & G_n(\mathbb{C}^\infty) \\ \downarrow & & \downarrow & & \parallel \\ S^1 & \longrightarrow & V_n(\mathbb{C}^\infty)/SU(n) & \longrightarrow & G_n(\mathbb{C}^\infty) \end{array}$$

given n -plane. One sees that the space $\tilde{G}_n(\mathbb{C}^\infty) = V_n(\mathbb{C}^\infty)/SU(n)$ is 2-connected by looking at the relevant portion of the diagram of homotopy groups associated to these two bundles:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \pi_2(G_n) & \xrightarrow[\approx]{\partial} & \pi_1(U(n)) & \longrightarrow & 0 \\ & & \parallel & & \downarrow \approx & & \\ 0 & \longrightarrow & \pi_2(\tilde{G}_n) & \longrightarrow & \pi_2(G_n) & \xrightarrow{\partial} & \pi_1(S^1) \longrightarrow \pi_1(\tilde{G}_n) \longrightarrow 0 \end{array}$$

The second vertical map is an isomorphism since S^1 embeds in $U(n)$ as the subgroup $U(1)$. Since the boundary map in the upper row is an isomorphism, so also is the boundary map in the lower row, and then exactness implies that $\tilde{G}_n$ is 2-connected.

The Gysin sequence for $S^1 \rightarrow \tilde{G}_n(\mathbb{C}^\infty) \rightarrow G_n(\mathbb{C}^\infty)$ can be analyzed just as in the preceding example. Part of the sequence is $H^0(G_n; \mathbb{Z}) \xrightarrow{\smile e} H^2(G_n; \mathbb{Z}) \rightarrow H^2(\tilde{G}_n; \mathbb{Z})$, and this last group is zero since $\tilde{G}_n$ is 2-connected, so e must be a generator of $H^2(G_n; \mathbb{Z}) \approx \mathbb{Z}$. Since $H^*(G_n; \mathbb{Z})$ is a polynomial algebra $\mathbb{Z}[c_1, \dots, c_n]$, we must have $e = \pm c_1$, so the map $\smile e: H^*(G_n; \mathbb{Z}) \rightarrow H^*(G_n; \mathbb{Z})$ is injective, the Gysin sequence breaks up into short exact sequences, and $H^*(\tilde{G}_n; \mathbb{Z})$ is the quotient ring $\mathbb{Z}[c_1, \dots, c_n]/(c_1) \approx \mathbb{Z}[c_2, \dots, c_n]$.

The spaces $\tilde{G}_n$ in the last two examples are often denoted $BSO(n)$ and $BSU(n)$, expressing the fact that they are related to the groups $SO(n)$ and $SU(n)$ via bundles $SO(n) \rightarrow V_n(\mathbb{R}^\infty) \rightarrow BSO(n)$ and $SU(n) \rightarrow V_n(\mathbb{C}^\infty) \rightarrow BSU(n)$ with contractible total spaces V_n . There is no quaternion analog of $BSO(n)$ and $BSU(n)$ since for $n = 2$ this would give a space with cohomology ring $\mathbb{Z}[x]$ on an 8-dimensional generator, which is impossible by Corollary 4L.10.

Now we turn to the derivation of the Gysin sequence, which follows a rather roundabout route:

- (1) Deduce a relative version of the Leray-Hirsch theorem from the absolute case.
- (2) Specialize this to the case of bundles with fiber a disk, yielding a basic result called the Thom isomorphism.
- (3) Show this applies to all orientable disk bundles.
- (4) Deduce the Gysin sequence by plugging the Thom isomorphism into the long exact sequence of cohomology groups for the pair consisting of a disk bundle and its boundary sphere bundle.

(1) A **fiber bundle pair** consists of a fiber bundle $p: E \rightarrow B$ with fiber F , together with a subspace $E' \subset E$ such that $p: E' \rightarrow B$ is a bundle with fiber a subspace $F' \subset F$, with local trivializations for E' obtained by restricting local trivializations for E . For example, if $E \rightarrow B$ is a bundle with fiber D^n and $E' \subset E$ is the union of the boundary spheres of the fibers, then (E, E') is a fiber bundle pair since local trivializations of E restrict to local trivializations of E' , in view of the fact that homeomorphisms from an n -disk to an n -disk restrict to homeomorphisms between their boundary spheres, boundary and interior points of D^n being distinguished by the local homology groups $H_n(D^n, D^n - \{x\}; \mathbb{Z})$.

Theorem 4D.8. Suppose that $(F, F') \rightarrow (E, E') \xrightarrow{p} B$ is a fiber bundle pair such that $H^*(F, F'; R)$ is a free R -module, finitely generated in each dimension. If there exist classes $c_j \in H^*(E, E'; R)$ whose restrictions form a basis for $H^*(F, F'; R)$ in each fiber (F, F') , then $H^*(E, E'; R)$, as a module over $H^*(B; R)$, is free with basis $\{c_j\}$.

The module structure is defined just as in the absolute case by $bc = p^*(b) \smile c$, but now we use the relative cup product $H^*(E; R) \times H^*(E, E'; R) \rightarrow H^*(E, E'; R)$.

Proof: Construct a bundle $\hat{E} \rightarrow B$ from E by attaching the mapping cylinder M of $p: E' \rightarrow B$ to E by identifying the subspaces $E' \subset E$ and $E' \subset M$. Thus the fibers $\hat{F}$ of $\hat{E}$ are obtained from the fibers F by attaching cones CF' on the subspaces $F' \subset F$. Regarding B as the subspace of $\hat{E}$ at one end of the mapping cylinder M , we have $H^*(\hat{E}, M; R) \approx H^*(\hat{E} - B, M - B; R) \approx H^*(E, E'; R)$ via excision and the obvious deformation retraction of $\hat{E} - B$ onto E . The long exact sequence of a triple gives $H^*(\hat{E}, M; R) \approx H^*(\hat{E}, B; R)$ since M deformation retracts to B . All these isomorphisms are $H^*(B; R)$ -module isomorphisms. Since B is a retract of $\hat{E}$ via the bundle projection $\hat{E} \rightarrow B$, we have a splitting $H^*(\hat{E}; R) \approx H^*(\hat{E}, B; R) \oplus H^*(B; R)$ as $H^*(B; R)$ -modules. Let $\hat{c}_j \in H^*(\hat{E}; R)$ correspond to $c_j \in H^*(E, E'; R) \approx H^*(\hat{E}, B; R)$ in this splitting. The classes $\hat{c}_j$ together with 1 restrict to a basis for $H^*(\hat{F}; R)$ in each fiber $\hat{F} = F \cup CF'$, so the absolute form of the Leray-Hirsch theorem implies that $H^*(\hat{E}; R)$ is a free $H^*(B; R)$ -module with basis $\{1, \hat{c}_j\}$. It follows that $\{c_j\}$ is a basis for the free $H^*(B; R)$ -module $H^*(E, E'; R)$. $\square$

(2) Now we specialize to the case of a fiber bundle pair $(D^n, S^{n-1}) \rightarrow (E, E') \xrightarrow{p} B$. An element $c \in H^n(E, E'; R)$ whose restriction to each fiber (D^n, S^{n-1}) is a generator of $H^n(D^n, S^{n-1}; R) \approx R$ is called a **Thom class** for the bundle. We are mainly interested in the cases $R = \mathbb{Z}$ and $\mathbb{Z}_2$, but R could be any commutative ring with identity, in which case a ‘generator’ is an element with a multiplicative inverse, so all elements of R are multiples of the generator. A Thom class with $\mathbb{Z}$ coefficients gives rise to a Thom class with any other coefficient ring R under the homomorphism $H^n(E, E'; \mathbb{Z}) \rightarrow H^n(E, E'; R)$ induced by the homomorphism $\mathbb{Z} \rightarrow R$ sending 1 to the identity element of R .

Corollary 4D.9. If the disk bundle $(D^n, S^{n-1}) \rightarrow (E, E') \xrightarrow{p} B$ has a Thom class $c \in H^n(E, E'; R)$, then the map $\Phi: H^i(B; R) \rightarrow H^{i+n}(E, E'; R)$, $\Phi(b) = p^*(b) \smile c$, is an isomorphism for all $i \geq 0$, and $H^i(E, E'; R) = 0$ for $i < n$. $\square$

The isomorphism Φ is called the **Thom isomorphism**. The corollary can be made into a statement about absolute cohomology by defining the **Thom space** $T(E)$ to be the quotient E/E' . Each disk fiber D^n of E becomes a sphere S^n in $T(E)$, and all these spheres coming from different fibers are disjoint except for the common base-point $x_0 = E'/E'$. A Thom class can be regarded as an element of $H^n(T(E), x_0; R) \approx$

$H^n(T(E); R)$ that restricts to a generator of $H^n(S^n; R)$ in each ‘fiber’ S^n in $T(E)$, and the Thom isomorphism becomes $H^i(B; R) \approx \tilde{H}^{n+i}(T(E); R)$.

(3) The major remaining step in the derivation of the Gysin sequence is to relate the existence of a Thom class for a disk bundle $D^n \rightarrow E \rightarrow B$ to a notion of orientability of the bundle. First we define orientability for a sphere bundle $S^{n-1} \rightarrow E' \rightarrow B$. In the proof of Proposition 4.61 we described a procedure for lifting paths γ in B to homotopy equivalences L_γ between the fibers above the endpoints of γ . We did this for fibrations rather than fiber bundles, but the method applies equally well to fiber bundles whose fiber is a CW complex since the homotopy lifting property was used only for the fiber and for the product of the fiber with I . In the case of a sphere bundle $S^{n-1} \rightarrow E' \rightarrow B$, if γ is a loop in B then L_γ is a homotopy equivalence from the fiber S^{n-1} over the basepoint of γ to itself, and we define the sphere bundle to be **orientable** if L_γ induces the identity map on $H^{n-1}(S^{n-1}; \mathbb{Z})$ for each loop γ in B .

For example, the Klein bottle, regarded as a bundle over S^1 with fiber S^1 , is nonorientable since as we follow a path looping once around the base circle, the corresponding fiber circles sweep out the full Klein bottle, ending up where they started but with orientation reversed. The same reasoning shows that the torus, viewed as a circle bundle over S^1 , is orientable. More generally, any sphere bundle that is a product is orientable since the maps L_γ can be taken to be the identity for all loops γ . Also, sphere bundles over simply-connected base spaces are orientable since $\gamma \simeq \eta$ implies $L_\gamma \simeq L_\eta$, hence all L_γ ’s are homotopic to the identity when all loops γ are nullhomotopic.

One could define orientability for a disk bundle $D^n \rightarrow E \rightarrow B$ by relativizing the previous definition, constructing lifts L_γ which are homotopy equivalences of the fiber pairs (D^n, S^{n-1}) . However, since $H^n(D^n, S^{n-1}; \mathbb{Z})$ is canonically isomorphic to $H^{n-1}(S^{n-1}; \mathbb{Z})$ via the coboundary map in the long exact sequence of the pair, it is simpler and amounts to the same thing just to define E to be orientable if its boundary sphere subbundle E' is orientable.

Theorem 4D.10. *Every disk bundle has a Thom class with $\mathbb{Z}_2$ coefficients, and orientable disk bundles have Thom classes with $\mathbb{Z}$ coefficients.*

An exercise at the end of the section is to show that the converse of the last statement is also true: A disk bundle is orientable if it has a Thom class with $\mathbb{Z}$ coefficients.

Proof: The case of a non-CW base space B reduces to the CW case by pulling back over a CW approximation to B , as in the Leray-Hirsch theorem, applying the five-lemma to say that the pullback bundle has isomorphic homotopy groups, hence isomorphic absolute and relative cohomology groups. From the definition of the pullback bundle it is immediate that the pullback of an orientable sphere bundle is orientable. There is also no harm in assuming the base CW complex B is connected. We will show:

If the disk bundle $D^n \rightarrow E \rightarrow B$ is orientable and B is a connected CW complex, (*) then the restriction map $H^i(E, E'; \mathbb{Z}) \rightarrow H^i(D_x^n, S_x^{n-1}; \mathbb{Z})$ is an isomorphism for all fibers D_x^n , $x \in B$, and for all $i \leq n$.

For $\mathbb{Z}_2$ coefficients we will see that (*) holds without any orientability hypothesis. Hence with either $\mathbb{Z}$ or $\mathbb{Z}_2$ coefficients, a generator of $H^n(E, E') \approx H^n(D_x^n, S_x^{n-1})$ is a Thom class.

If the disk bundle $D^n \rightarrow E \rightarrow B$ is orientable, then if we choose an isomorphism $H^n(D_x^n, S_x^{n-1}; \mathbb{Z}) \approx \mathbb{Z}$ for one fiber D_x^n , this determines such isomorphisms for all fibers by composing with the isomorphisms L_y^* , which depend only on the endpoints of y . Having made such a choice, then if (*) is true, we have a preferred isomorphism $H^n(E, E'; \mathbb{Z}) \approx \mathbb{Z}$ which restricts to the chosen isomorphism $H^n(D_x^n, S_x^{n-1}; \mathbb{Z}) \approx \mathbb{Z}$ for each fiber. This is because for a path y from x to y , the inclusion $(D_x^n, S_x^{n-1}) \hookrightarrow (E, E')$ is homotopic to the composition of L_y with the inclusion $(D_y^n, S_y^{n-1}) \hookrightarrow (E, E')$. We will use this preferred isomorphism $H^n(E, E'; \mathbb{Z}) \approx \mathbb{Z}$ in the inductive proof of (*) given below. In the case of $\mathbb{Z}_2$ coefficients, there can be only one isomorphism of a group with $\mathbb{Z}_2$ so no choices are necessary and orientability is irrelevant. We will prove (*) in the $\mathbb{Z}$ coefficient case, leaving it to the reader to replace all $\mathbb{Z}$'s in the proof by $\mathbb{Z}_2$'s to obtain a proof in the $\mathbb{Z}_2$ case.

Suppose first that the CW complex B has finite dimension k . Let $U \subset B$ be the subspace obtained by deleting one point from the interior of each k -cell of B , and let $V \subset B$ be the union of the open k -cells. Thus $B = U \cup V$. For a subspace $A \subset B$ let $E_A \rightarrow A$ and $E'_A \rightarrow A$ be the disk and sphere bundles obtained by taking the subspaces of E and E' projecting to A . Consider the following portion of a Mayer-Vietoris sequence, with $\mathbb{Z}$ coefficients implicit from now on:

$$H^n(E, E') \rightarrow H^n(E_U, E'_U) \oplus H^n(E_V, E'_V) \xrightarrow{\Psi} H^n(E_{U \cap V}, E'_{U \cap V})$$

The first map is injective since the preceding term in the sequence is zero by induction on k , since $U \cap V$ deformation retracts onto a disjoint union of $(k-1)$ -spheres and we can apply Lemma 4D.2 to replace $E_{U \cap V}$ by the part of E over this union of $(k-1)$ -spheres. By exactness we then have an isomorphism $H^n(E, E') \approx \text{Ker } \Psi$. Similarly, by Lemma 4D.2 and induction each of the terms $H^n(E_U, E'_U)$, $H^n(E_V, E'_V)$, and $H^n(E_{U \cap V}, E'_{U \cap V})$ is a product of $\mathbb{Z}$'s, with one $\mathbb{Z}$ factor for each component of the spaces involved, projection onto the $\mathbb{Z}$ factor being given by restriction to any fiber in the component. Elements of $\text{Ker } \Psi$ are pairs $(\alpha, \beta) \in H^n(E_U, E'_U) \oplus H^n(E_V, E'_V)$ having the same restriction to $H^n(E_{U \cap V}, E'_{U \cap V})$. Since B is connected, this means that all the $\mathbb{Z}$ coordinates of α and β in the previous direct product decompositions must be equal, since between any two components of U or V one can interpolate a finite sequence of components of U and V alternately, each component in the sequence having nontrivial intersection with its neighbors. Thus $\text{Ker } \Psi$ is a copy of $\mathbb{Z}$, with restriction to a fiber being the isomorphism $H^n(E, E') \approx \mathbb{Z}$.

To finish proving $(*)$ for finite-dimensional B it remains to see that $H^i(E, E') = 0$ for $i < n$, but this follows immediately by looking at an earlier stage of the Mayer-Vietoris sequence, where the two terms adjacent to $H^i(E, E')$ vanish by induction.

Proving $(*)$ for an infinite-dimensional CW complex B reduces to the finite-dimensional case as in the Leray-Hirsch theorem since we are only interested in cohomology in a finite range of dimensions. $\square$

(4) Now we can derive the Gysin sequence for a sphere bundle $S^{n-1} \rightarrow E \xrightarrow{p} B$. Consider the mapping cylinder M_p , which is a disk bundle $D^n \rightarrow M_p \xrightarrow{p} B$ with E as its boundary sphere bundle. Assuming that a Thom class $c \in H^n(M_p, E; R)$ exists, as is the case if E is orientable or if $R = \mathbb{Z}_2$, then the long exact sequence of cohomology groups for the pair (M_p, E) gives the first row of the following commutative diagram, with R coefficients implicit:

$$\begin{array}{ccccccc} \cdots & \longrightarrow & H^i(M_p, E) & \xrightarrow{j^*} & H^i(M_p) & \longrightarrow & H^i(E) \longrightarrow H^{i+1}(M_p, E) \longrightarrow \cdots \\ & & \approx \uparrow \Phi & & \approx \uparrow p^* & & \parallel \\ \cdots & \longrightarrow & H^{i-n}(B) & \xrightarrow{\smile e} & H^i(B) & \xrightarrow{p^*} & H^i(E) \longrightarrow H^{i-n+1}(B) \longrightarrow \cdots \end{array}$$

The maps Φ are the Thom isomorphism, and the vertical map p^* is an isomorphism since M_p deformation retracts onto B . The **Euler class** $e \in H^n(B; R)$ is defined to be $(p^*)^{-1}j^*(c)$, c being a Thom class. The square containing the map $\smile e$ commutes since for $b \in H^{i-n}(B; R)$ we have $j^*\Phi(b) = j^*(p^*(b) \smile c) = p^*(b) \smile j^*(c)$, which equals $p^*(b \smile e) = p^*(b) \smile p^*(e)$ since $p^*(e) = j^*(c)$. Another way of defining e is as the class corresponding to $c \smile c$ under the Thom isomorphism, since $\Phi(e) = p^*(e) \smile c = j^*(c) \smile c = c \smile c$.

Finally, the lower row of the diagram is by definition the Gysin sequence. $\square$

To conclude this section we will use the following rather specialized application of the Gysin sequence to compute a few more examples of spaces with polynomial cohomology.

Proposition 4D.11. *Suppose that $S^{2k-1} \rightarrow E \xrightarrow{p} B$ is an orientable sphere bundle such that $H^*(E; R)$ is a polynomial ring $R[x_1, \dots, x_\ell]$ on even-dimensional generators x_i . Then $H^*(B; R) = R[y_1, \dots, y_\ell, e]$ where e is the Euler class of the bundle and $p^*(y_i) = x_i$ for each i .*

Proof: Consider the three terms $H^i(B; R) \xrightarrow{\smile e} H^{i+2k}(B; R) \longrightarrow H^{i+2k}(E; R)$ of the Gysin sequence. If i is odd, the third term is zero since E has no odd-dimensional cohomology. Hence the map $\smile e$ is surjective, and by induction on dimension this implies that $H^*(B; R)$ is zero in odd dimensions. This means the Gysin sequence reduces to short exact sequences

$$0 \longrightarrow H^{2i}(B; R) \xrightarrow{\smile e} H^{2i+2k}(B; R) \xrightarrow{p^*} H^{2i+2k}(E; R) \longrightarrow 0$$

Since p^* is surjective, we can choose elements $y_j \in H^*(B; R)$ with $p^*(y_j) = x_j$. It remains to check that $H^*(B; R) = R[y_1, \dots, y_\ell, e]$, which is elementary algebra: Given $b \in H^*(B; R)$, $p^*(b)$ must be a polynomial $f(x_1, \dots, x_\ell)$, so $b - f(y_1, \dots, y_\ell)$ is in the kernel of p^* and exactness gives an equation $b - f(y_1, \dots, y_\ell) = b' \smile e$ for some $b' \in H^*(B; R)$. Since b' has lower dimension than b , we may assume by induction that b' is a polynomial in $y_1, \dots, y_\ell, e$. Hence $b = f(y_1, \dots, y_\ell) + b' \smile e$ is also a polynomial in $y_1, \dots, y_\ell, e$. Thus the natural map $R[y_1, \dots, y_\ell, e] \rightarrow H^*(B; R)$ is surjective. To see that it is injective, suppose there is a polynomial relation $f(y_1, \dots, y_\ell, e) = 0$ in $H^*(B; R)$. Applying p^* , we get $f(x_1, \dots, x_\ell, 0) = 0$ since $p^*(y_i) = x_i$ and $p^*(e) = 0$ from the short exact sequence. The relation $f(x_1, \dots, x_\ell, 0) = 0$ takes place in the polynomial ring $R[x_1, \dots, x_\ell]$, so $f(y_1, \dots, y_\ell, 0) = 0$ in $R[y_1, \dots, y_\ell, e]$, hence $f(y_1, \dots, y_\ell, e)$ must be divisible by e , say $f = ge$ for some polynomial g . The relation $f(y_1, \dots, y_\ell, e) = 0$ in $H^*(B; R)$ then has the form $g(y_1, \dots, y_\ell, e) \smile e = 0$. Since $\smile e$ is injective, this gives a polynomial relation $g(y_1, \dots, y_\ell, e) = 0$ with g having lower degree than f . By induction we deduce that g must be the zero polynomial, hence also f . $\square$

Example 4D.12. Let us apply this to give another proof that $H^*(G_n(\mathbb{C}^\infty); \mathbb{Z})$ is a polynomial ring $\mathbb{Z}[c_1, \dots, c_n]$ with $|c_i| = 2i$. We use two fiber bundles:

$$S^{2n-1} \rightarrow E \rightarrow G_n(\mathbb{C}^\infty) \quad S^\infty \rightarrow E \rightarrow G_{n-1}(\mathbb{C}^\infty)$$

The total space E in both cases is the space of pairs (P, v) where P is an n -plane in $\mathbb{C}^\infty$ and v is a unit vector in P . In the first bundle the map $E \rightarrow G_n(\mathbb{C}^\infty)$ is $(P, v) \mapsto P$, with fiber S^{2n-1} , and for the second bundle the map $E \rightarrow G_{n-1}(\mathbb{C}^\infty)$ sends (P, v) to the $(n-1)$ -plane in P orthogonal to v , with fiber S^∞ consisting of all the unit vectors in $\mathbb{C}^\infty$ orthogonal to a given $(n-1)$ -plane. Local triviality for the two bundles is verified in the usual way. Since S^∞ is contractible, the map $E \rightarrow G_{n-1}(\mathbb{C}^\infty)$ induces isomorphisms on all homotopy groups, hence also on all cohomology groups. By induction on n we then have $H^*(E; \mathbb{Z}) \approx \mathbb{Z}[c_1, \dots, c_{n-1}]$. The first bundle is orientable since $G_n(\mathbb{C}^\infty)$ is simply-connected, so the corollary gives $H^*(G_n(\mathbb{C}^\infty); \mathbb{Z}) \approx \mathbb{Z}[c_1, \dots, c_n]$ with $c_n = e$.

The same argument works in the quaternionic case. For a version of this argument in the real case see §3.3 of [VBKT].

Before giving our next example, let us observe that the Gysin sequence with a fixed coefficient ring R is valid for any orientable fiber bundle $F \rightarrow E \xrightarrow{p} B$ whose fiber is a CW complex F with $H^*(F; R) \approx H^*(S^{n-1}; R)$. Orientability is defined just as before in terms of induced maps $L_y^*: H^{n-1}(F; R) \rightarrow H^{n-1}(F; R)$. No changes are needed in the derivation of the Gysin sequence to get this more general case, if the associated ‘disk’ bundle is again taken to be the mapping cylinder $CF \rightarrow M_p \rightarrow B$.

Example 4D.13. We have computed the cohomology of $\tilde{G}_n(\mathbb{R}^\infty)$ with $\mathbb{Z}_2$ coefficients, finding it to be a polynomial ring on generators in dimensions 2 through n , and now